Molecular representations, similarity and molecular descriptors

Molecular representation –why it matters

- **Molecules as data**
  - Computers need structured data
  - Molecular representations translate chemistry into numbers, graphs and strings
  - Algorithms can reason about chemical space computationally

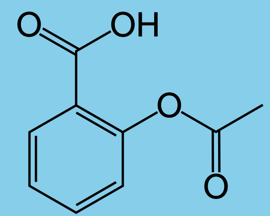
- **2D- versus 3D representations**
  - 2D – connectivity and bonding (SMILES, InChI)
  - 3D – spatial geometry, conformation, chirality (pdb, sdf, mol2)
  - Drug activity often depends **shape** and **orientation** – not just formula

- **Standardization and interoperability**
  - Formats (SDF, MOl2, PDB, SMILES) ensure compatibility across software
  - Reproducibility depends on **consistent molecular definitions** (e.g. tautomers, protonation states
  - Critical for automated workflows like virtual screening and QSAR

- **From representation to prediction**
  - The quality of the model depends on how well it represents reality
  - Proper encoding needed for **docking**, **scoring,** and **ML-based property prediction**
  - “Garbage in, garbage out” — accurate representations are the foundation of reliable results.

2D molecular representation

$C_9H_8O_4$ O=C(C)Oc1ccccc1C(=O)O



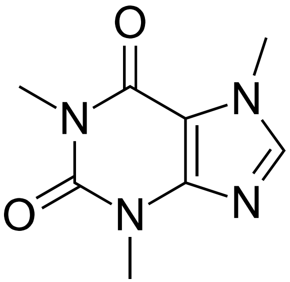
InChI=1S/C9H8O4/c1-6(10)13-8-5-3-2-4-7(8)9(11)12/h2-5H,1H3,(H,11,12)

Key:BSYNRYMUTXBXSQ-UHFFFAOYSA-N

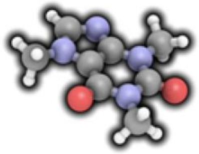
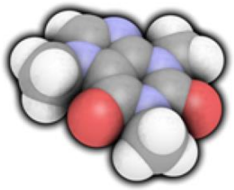
- **Abstract the molecule's topology**
  - Capture how atoms are connected through bonds, without explicit 3D coordinates
- **Serve as the foundation for cheminformatics**
  - Used in databases, QSAR models and similarity searches
- **Common formats**
  - SMILES, InChI, and connection tables (as in MOL or SDF files).
- **Efficient for computation**
  - enable fast storage, indexing, and comparison of large compound libraries.
- **Limitations**
  - 2D formats can capture spatial arrangements, conformations and geometry information like bond lengths, bond angles, etc..

Representing chemical structures

- Exampe: representing the molecule of caffeine in different ways



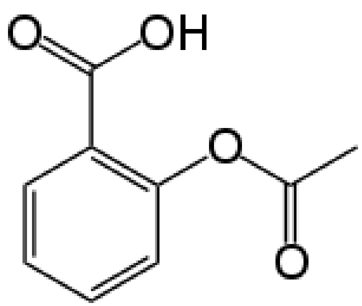
Representation Name	Representation of Caffeine
Common Name	Caffeine
Synonyms	Guaranine
	Methyltheobromine
	1,3,7-Trimethylxanthine
	Theine
Empirical Formula	$C_8H_{10}N_4O_2$
IUPAC Name	1,3,7-trimethylpurine-2,6-dione
CAS Registry Number	58-08-2
ChEMBL ID	CHEMBL113
Wiswesser Line Notation (WLN)	T56 BN DN FNVNVJ B F H
SMILES	CN1C=NC2=C1C(=O)N(C(=O)N2C)C
Aromatic SMILES	CN1C(=O)N(C)c2ncn(C)22C1=O

Representation Name	Representation of Caffeine
InChI	1S/C8H10N4O2/c1-10-4-9-6-5(10)7(13)12(3)8(14)11(6)2/h4H,1-3H3
InChIKey	RYYVLZVUVIJVGH-UHFFFAOYSA-N
Topography	
Surface	

SMILES (1)

SMILES - Simplified Input Molecular Entry Specification

Linguagem que permite a representação de estruturas moleculares 2D na forma de uma sequência (“string”) de caracteres.



Estrutura 2D

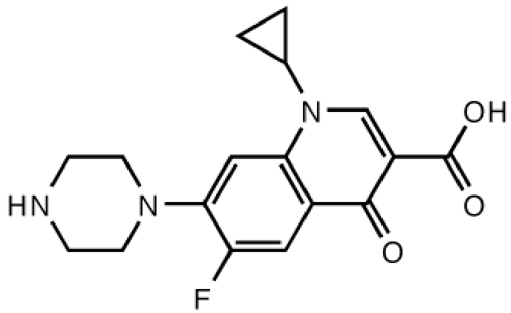
O=C(Oc1ccccc1C(=O)O)C

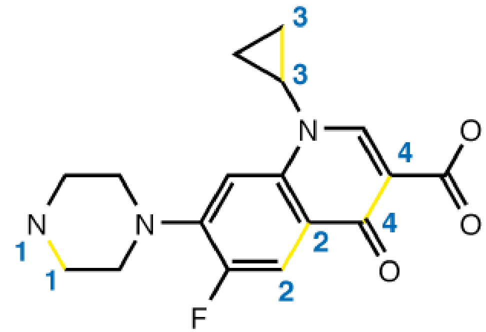
SMILES

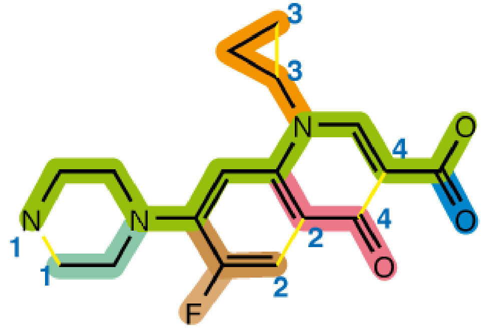
Tutorial SMILES: http://www.daylight.com/

D. Weininger (1988) *J. Chem. Inf. Comput. Sci.* **28**:31

SMILES (2)





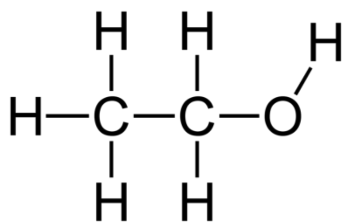


1. Chose the longest bond path
2. All other paths are branches ()
3. Rings are open, and atoms by cut bonds numbered
4. Double bonds are represent by “=“, single bonds implicit
5. Hydrogens are (generally) implict

N1CCN(CC1)C(C(F)=C2)=CC(=C2C4=O)N(C3CC3)C=C4C(=O)O

Canonical SMILES

- SMILES strings are not uniquely defined for each molecule, for example:



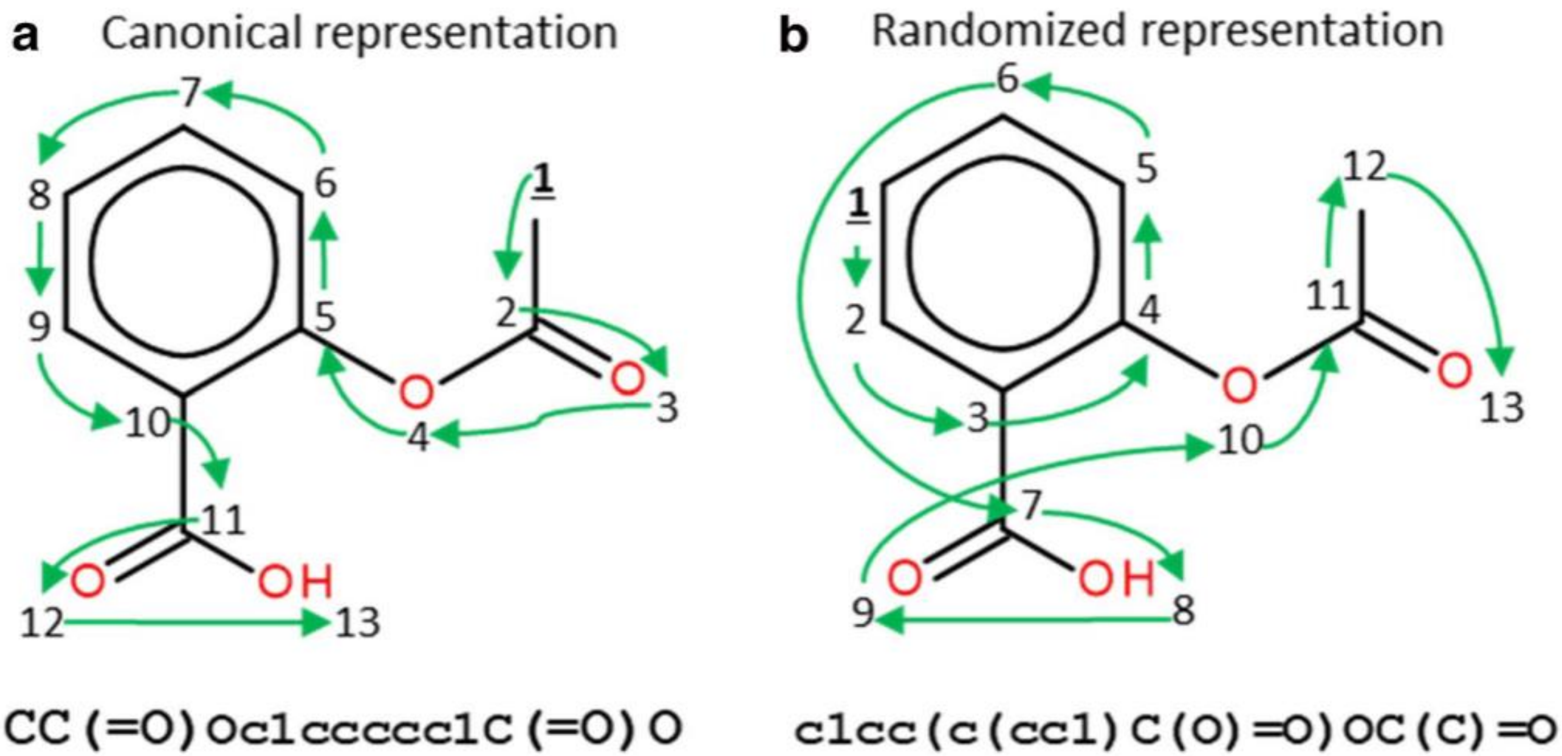
ethanol

CCO OCC C(O)C

All valid SMILES representations

- To solve this problem, *canonization algorithms* that produce a unique SMILES string for each structure have been developed
- Many software packages can generate canonical SMILES:
- Daylight Chemical Information Systems
- OpenEye Scientific Software
- Chemical Computing Group
- Chemistry Development Kit

Canonical versus randomized SMILES



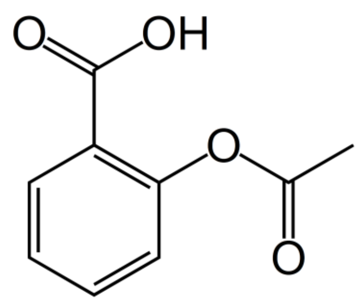
InChI representation

InChI – IUPAC International Chemical Identifier
Developed by IUPAC and NIST 2000-2005 (pronounced “In Key”)

InChI is a text-based identifier for chemical substances, designed to offer a standard way to provide molecular information

The InChI Identifier describes molecules in terms of different layers of information:

- Main Layer
  - Chemical Formula
  - Atom connections
  - Hydrogen Atoms
- Charge Layer
- Stereochemical Layer
- Isotopic Layer
- Fixed-H Layer
- Reconnected layer



InChI=1S/C9H8O4/c1-6(10)13-8-5-3-2-4-7(8)9(11)12/h2-5H,1H3,(H,11,12)

1-version number
S-standardized InChI

Chemical Formula

Connectivity

Hydrogen Atoms

InChI - Rifampicin

InChI=1S/C43H58N4O12/c1-21-12-11-13-22(2)42(55)45-33-28(20-44-47-17-15-46(9)16-18-47)37(52)30-31(38(33)53)36(51)26(6)40-32(30)41(54)43(8,59-40)57-19-14-29(56-10)23(3)39(58-27(7)48)25(5)35(50)24(4)34(21)49/h11-14,19-21,23-25,29,34-35,39,49-53H,15-18H2,1-10H3,(H,45,55)/b12-11+,19-14+,22-13-,44-20?/t21-,23+,24+,25+,29-,34-,35+,39+,43-/m0/s1

Chemical Forumula - $C_{43}H_{58}N_4O_{12}$

Connectivity - c1-21-12-11-13...

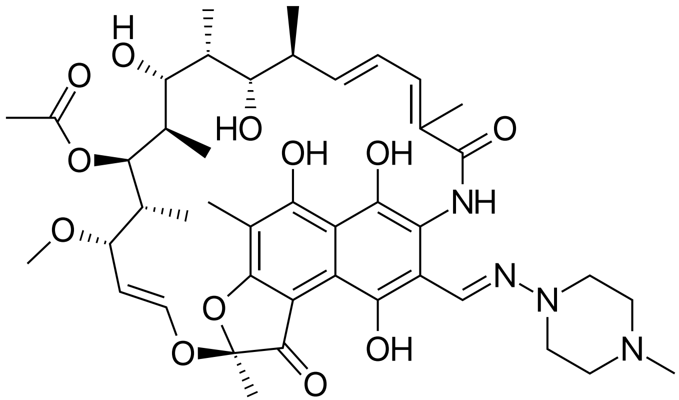
Hydrogem atoms - h11-14,19-21,23-25,29,3...

Double bond stereochemistry - b12-11+,19-14+,22-13-...

Tetrahedral stereochemistry - t21-,23+,24+,25+,29-,34-...

m0 – absolute stereochemical configuration

S1 – no inversion



- Ansamycin antibiotic with complex structure
- 9 chiral centers
- 4 double bonds with E/Z configurations
- Hetero-ring systems

InChI is canonical by default

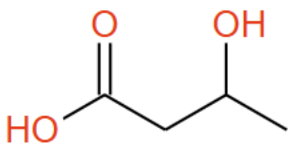
- For InChi generation, atoms are numbered according to a canonical algorithm that ensures unique sequence for a given structures

Same molecule:

CC(O)CC(=O)O InChI=1S/C4H8O3/c1-3(5)2-4(6)7/h3,5H,2H2,1H3,(H,6,7)

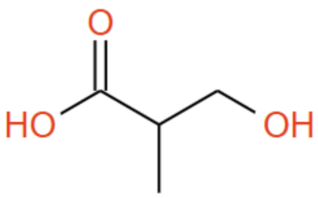
C(C(C)O)C(=O)O InChI=1S/C4H8O3/c1-3(5)2-4(6)7/h3,5H,2H2,1H3,(H,6,7)

C(CC(=O)O)(C)O InChI=1S/C4H8O3/c1-3(5)2-4(6)7/h3,5H,2H2,1H3,(H,6,7)



Differnet molecle:

OCC(C)C(=O)O InChI=1S/C4H8O3/c1-3(2-5)4(6)7/h3,5H,2H2,1H3,(H,6,7)



InChI key

InChI's are too long and complex to reliably work as search keywords in database/internet searches.

InChI Key – Compressed form of the InChI, using a hashing algorithm (sha-256) to produce a quasi-unique alphabetic string with shorter length.

Different InChI's can produce the same InChI Key, but that's an extremely rare event. The **connectivity** layer is hashed separately – two molecules with the same connectivity but with different spatial arrangement will have the same first part of InChI Key

InChI=1S/C9H8O4/c1-6(10)13-8-5-3-2-4-7(8)9(11)12/h2-5H,1H3,(H,11,12)

sha-256

InChIKey BSYNRYMUTXBXSQ-UHFFFAOYSA-N

Connectivity (BSYNRYMUTXBXSQ)

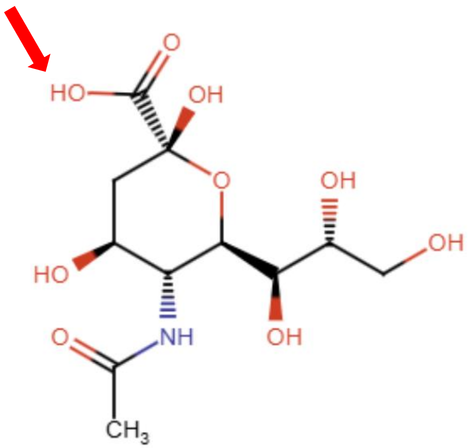
Other layers (UHFFFAOY)

Standard /Non-standard (S)

Version (A)

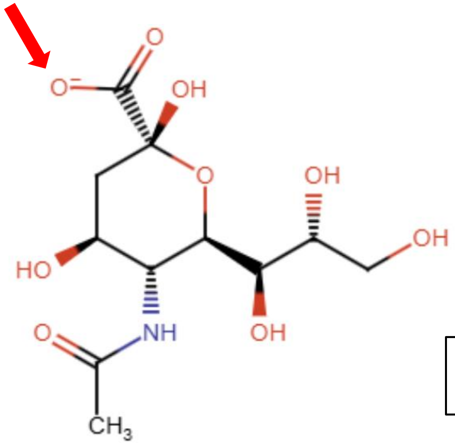
Protonation state (N)

InChI key – sialic acid



N-acetyl-alpha-neuraminic acid

SQVRNKJHWKZAKO-YRMXFSIDSA-N

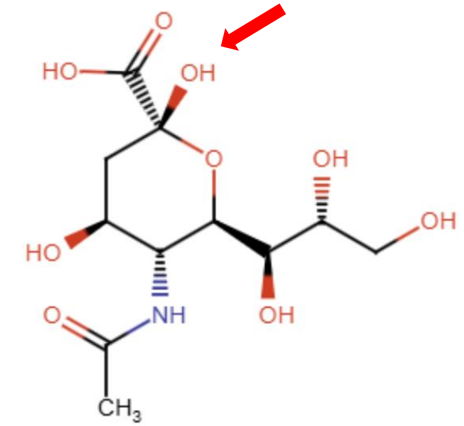


N-acetyl-alpha-neuraminate

SQVRNKJHWKZAKO-YRMXFSIDSA-M

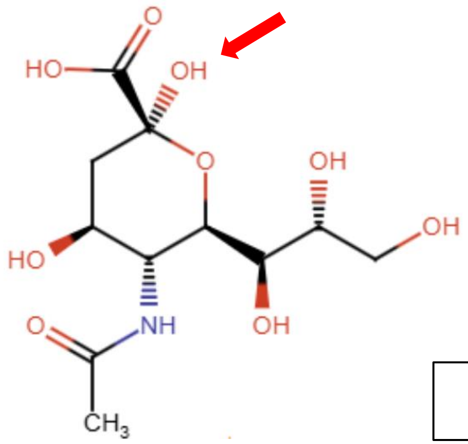
Change is only in **protonation state**

InChI key – sialic acid



N-acetyl-alpha-neuraminic acid

SQVRNKJHWKZAKO-YRMXFSIDSA-N



N-acetyl-beta-neuraminic acid

SQVRNKJHWKZAKO-PFQGKNLYSA-N

Change is only in stereo-isomerism layer of InChI

Online InChI(Key) generator

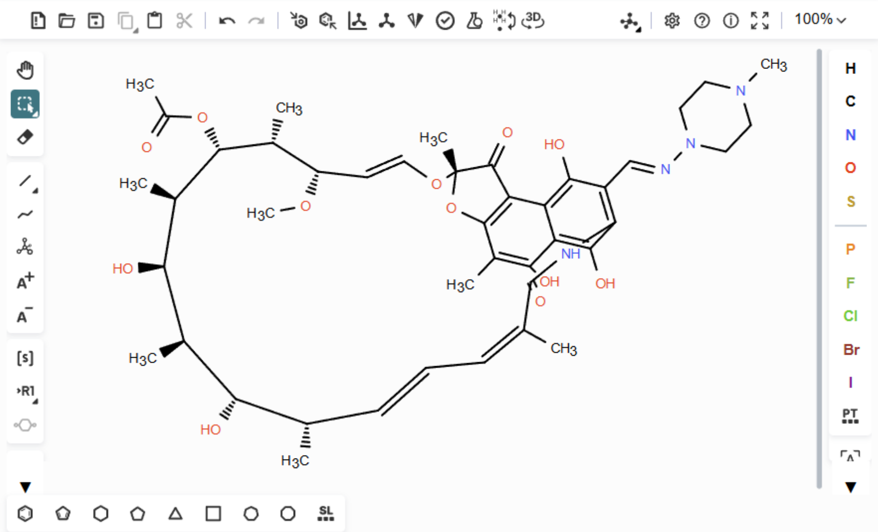
InChI TRUST

InChI Web Demo

https://iupac-inchi.github.io/InChI-Web-Demo/

InChI RInChI About

Draw structure and convert to InChI | Convert Molfile to InChI | Convert AuxInfo to structure | Convert SDF to InChIs



Version

Latest

https://github.com/IUPAC-InChI/InChI/releases/tag/v1.07.4

Options

Tautomer options

- [x] Mobile H Perception
- [ ] Do not add implicit H atoms
- [x] Include Stereo: (•) Absolute ( ) Relative ( ) Racemic ( ) From chiral flag
  - [ ] Always include omitted/undefined stereo
  - [ ] Different marks for unknown/undefined stereo
  - [ ] Both ends of wedge point to stereocenters
- [ ] Include Bonds to Metal
- [ ] Treat polymers:
  - [ ] No pre-edits of original polymer structure
  - [ ] Enable CRU folding
  - [ ] Disable CRU frame shift
- [ ] Allow non-polymer Zz pseudoatoms

Reset InChI Options

InChI

```
InChI=1S/C43H58N4O12/c1-21-12-11-13-22(2)42(55)45-33-28(20-44-47-17-15-46(9)16-18-47)37(52)30-31(38(33)53)36(51)26(6)40-32(30)41(54)43(8,59-40)57-19-14-29(56-10)23(3)39(58-27(7)48)25(5)35(50)24(4)34(21)49/h11-14,19-21,23-25,29,34-35,39,49-53H,15-18H2,1-10H3,(H,45,55)/b12-11+,19-14+,22-13-,44-20+/t21-,23+,24+,25+,29-,34-,35+,39+,43-/m0/s1
```

InChIKey

```
JQXXHWHPUNPDRT-WLSIYKJHSA-N
```

The importance of 3D representations

- **Capture the true shape of molecules:** 3D coordinates define geometry, orientation, and chirality, essential for biological activity.
- **Enable structure-based methods:** Docking, molecular dynamics, and pharmacophore modeling all rely on accurate 3D structures.
- **Bridge experimental and computational worlds:** Derived from crystallography (PDB, CIF) or built computationally (RDKit, Open Babel).
- **Determine molecular interactions:**3D structure governs binding affinity, selectivity, and conformational flexibility.

Common 3D representation formats

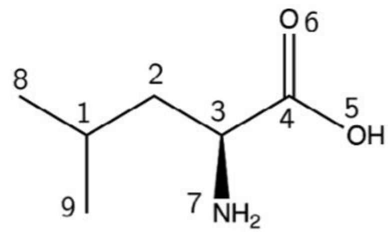
- **PDB (.pdb) :**
  - Standard format for macromolecules (proteins, nucleic acids, complexes); includes atomic coordinates and metadata such as chains, residues, and secondary structure.
- **MOL2 (.mol2) :**
  - Tripos format supporting atom types, charges, and bonding information; widely used for docking and molecular mechanics.
- **SDF (.sdf) :**
  - Structure Data File; can store multiple molecules with associated properties (e.g., activity data, ID numbers).
- **XYZ (.xyz) :**
  - Simple Cartesian coordinate format; easy to read, lacks bonding information.
- **CIF (.cif) :**
  - Crystallographic Information File; used for crystal structures and symmetry information (mainly from X-ray data).
- **PQR (.pqr)** :
  - Derived from PDB, includes partial atomic charges and radii (for for electrostatics calculations like APBS)
- **PDBQT(.pdbqt):**
  - Derived from PDB, includes atom types, charges and free torsions (used by docking programs like Autodock and Autodock Vina)

Structure format comparison

Format	Meaning / Origin	Typical Use	Key Features	Notes / Limitations
MOL (.mol)	MDL Molfile	Small molecules, editors like ChemDraw or RDKit	Single molecule; atoms, bonds, 2D/3D coordinates	No metadata or multiple structures
SDF (.sdf)	Structure Data File (MDL)	Compound libraries, QSAR datasets	Multiple MOL blocks + properties (activity, ID, etc.)	No atom types or charges; not ideal for simulations
MOL2 (.mol2)	Tripos Mol2 (SYBYL)	Docking, molecular mechanics	Atom types, charges, substructures, residues	More detailed, but less standardized across tools
PDB (.pdb)	Protein Data Bank format	Proteins, nucleic acids, large complexes	3D atomic coordinates, chains, residues, occupancy	Legacy format, limited precision (fixed columns)
CIF (.cif)	Crystallographic Information File	X-ray / neutron crystal structures	Unit cell, symmetry, atomic positions, metadata	Text-based, rich but complex (mmCIF used for macromolecules)

MDL mol file

Leucine



Chiral

```
  9  8  0  0  1  0  0  0  0  0999 V2000
   -1.0717   -0.2062    0.0000 C   0  0  0  0  0  0  0  0  0  0  0  0
   -0.3572    0.2062    0.0000 C   0  0  0  0  0  0  0  0  0  0  0  0
    0.3572   -0.2062    0.0000 C   0  0  0  0  0  0  0  0  0  0  0  0
    1.0717    0.2062    0.0000 C   0  0  0  0  0  0  0  0  0  0  0  0
    1.7861   -0.2062    0.0000 O   0  0  0  0  0  0  0  0  0  0  0  0
    1.0717    1.0312    0.0000 O   0  0  0  0  0  0  0  0  0  0  0  0
    0.3572   -1.0312    0.0000 N   0  0  0  0  0  0  0  0  0  0  0  0
   -1.7861    0.2062    0.0000 C   0  0  0  0  0  0  0  0  0  0  0  0
   -1.0717   -1.0312    0.0000 C   0  0  0  0  0  0  0  0  0  0  0  0
  1  2  1  0
  2  3  1  0
  3  4  1  0
  4  5  1  0
  4  6  2  0
  3  7  1  1
  1  8  1  0
  1  9  1  0
M  END
```

Counts Line

Atom Block

Bond Block

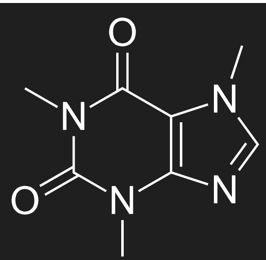
M END

Blocks not used in this Ctab:
Atom List, Stext, Properties

Trypos mol2 format

```
@<TRIPOS>MOLECULE
CAFINE
    24    25     1     0     0
SMALL
NO_CHARGES
****
Generated from the CSD

@<TRIPOS>ATOM
      1 C1       3.6210   3.7158  -0.3933  C.2      1 RES1    0.0000
      2 C2       1.4218   4.2301   0.5103  C.2      1 RES1    0.0000
      3 C3       1.1506   2.9375   0.7660  C.2      1 RES1    0.0000
      4 C4       2.1094   1.9088   0.4551  C.2      1 RES1    0.0000
      5 C5      -0.4705   4.2084   1.4335  C.2      1 RES1    0.0000
      6 C6       4.3372   1.3894  -0.4768  C.3      1 RES1    0.0000
      7 C7       2.9376   6.0755  -0.3117  C.3      1 RES1    0.0000
      8 C8      -0.9085   1.7585   1.8063  C.3      1 RES1    0.0000
      9 N1       3.2629   2.3631  -0.1044  N.am     1 RES1    0.0000
     10 N2       2.6728   4.6242  -0.0599  N.am     1 RES1    0.0000
     11 N3      -0.1337   2.9208   1.3303  N.pl3    1 RES1    0.0000
     12 N4       0.4784   5.0234   0.9615  N.2      1 RES1    0.0000
     13 O1       4.6487   4.0080  -0.9402  O.2      1 RES1    0.0000
     14 O2       1.9391   0.6747   0.6368  O.2      1 RES1    0.0000
     15 H1      -1.5169   4.3587   1.8678  H        1 RES1    0.0000
     16 H2      -0.4822   1.0354   2.3603  H        1 RES1    0.0000
     17 H3      -1.0965   1.0521   1.0954  H        1 RES1    0.0000
     18 H4      -1.8007   2.2879   2.0096  H        1 RES1    0.0000
     19 H5       3.9616   6.0454  -0.5635  H        1 RES1    0.0000
     20 H6       3.3236   6.6132   0.4137  H        1 RES1    0.0000
     21 H7       2.2066   6.2959  -0.8551  H        1 RES1    0.0000
     22 H8       5.2607   1.6700  -0.8984  H        1 RES1    0.0000
     23 H9       4.4294   0.5511   0.0867  H        1 RES1    0.0000
     24 H10      3.9604   1.0020  -1.2767  H        1 RES1    0.0000
```



atom names

x y z

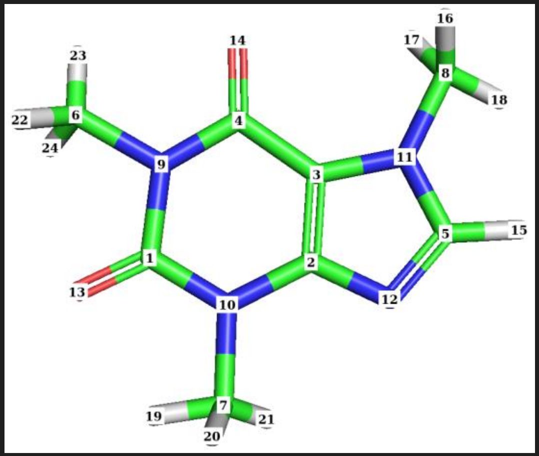
coordinates

atom types

- 3D representation including atoms, charges, atom types, bonds, and other information (.mol2 extension)
- Can be read by many software applications and libraries

bond types

```
@<TRIPOS>BOND
     1     1     9   am
     2     2     3    2
     3     3     4    1
     4     4     9   am
     5     5    11    1
     6     6     9    1
     7     7    10    1
     8     8    11    1
     9    10     1   am
    10    11     3    1
    11    12     2    1
    12    13     1    2
    13    14     4    2
    14    15     5    1
    15    16     8    1
    16    17     8    1
    17    18     8    1
    18    19     7    1
    19    20     7    1
    20    21     7    1
    21    22     6    1
    22    23     6    1
    23    24     6    1
    24     2    10    1
    25     5    12    2
@<TRIPOS>CRYSIN
   14.8000   16.7000    3.9700   90.0000   97.0000   90.0000    14     2
```



From 2D to 3D structure

3D structures may be obtained by different methods

- Experimental: X-ray crystallography, NMR, EPR, Cryo-EM and others
- Theoretical:
  - Quantum Chemstry – accurate, but slow. Limited conformation.
  - From 2D to 3D:
    - **Start with 2D input** — chemical drawings or SMILES encode atomic connectivity but lack spatial information
    - **Generate 3D conformers** — algorithms (e.g., RDKit ETKDG, Open Babel) build plausible 3D geometries from 2D graphs.
    - **Optimize geometry** — force fields (MMFF94, UFF) or quantum methods refine bond lengths and angles to realistic minima.
    - **Validate and export** — save optimized structures in 3D formats (SDF, MOL2, PDB) for use in docking or molecular dynamics.

The importance of molecular similarity

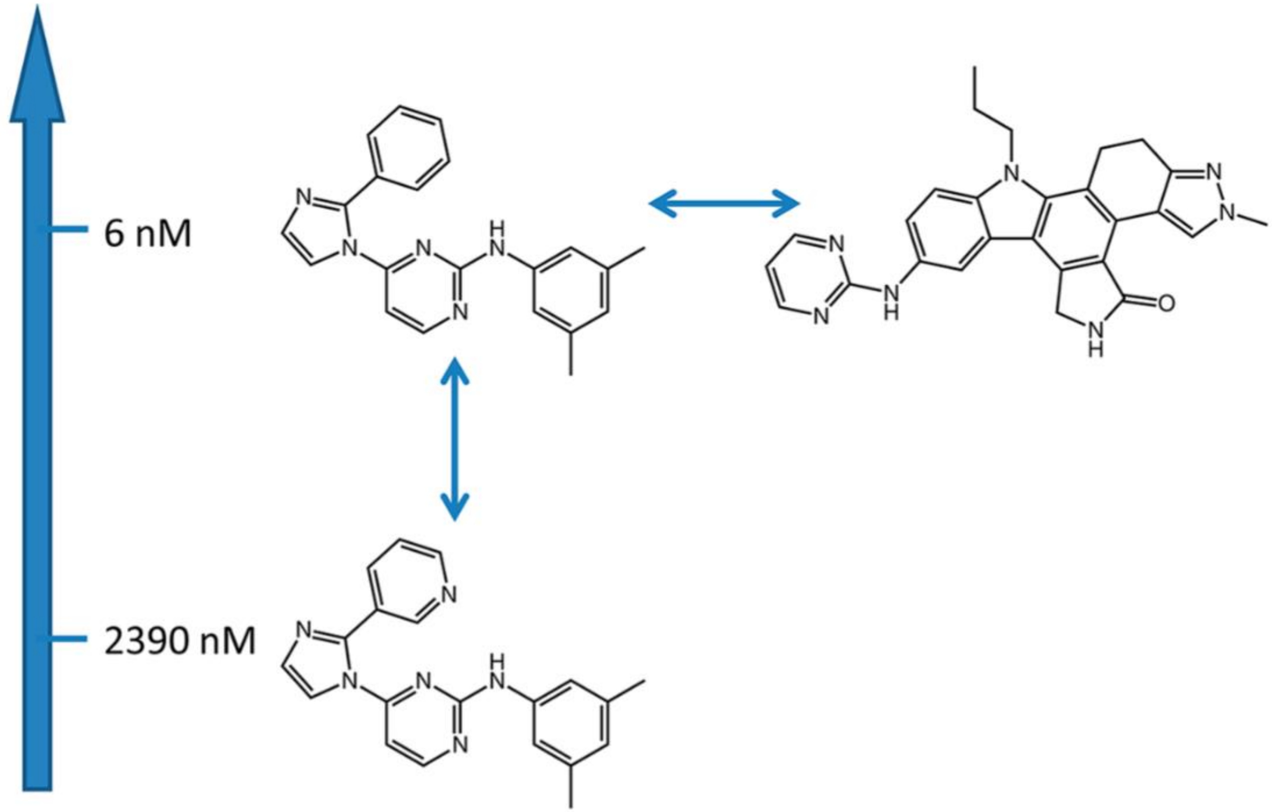
- **Concept and Motivation**
  - Similar molecules tend to have similar biological activity — the "similarity principle."
  - Central idea behind **ligand-based drug design**, **virtual screening**, and **QSAR**.
  - Helps find new hits when a target's structure is unknown.

- **How to Measure Similarity**
  - Compare **molecular fingerprints** or **descriptors** numerically.
  - Common metric: **Tanimoto coefficient** (0 = different, 1 = identical).
  - Alternative metrics: Dice, Cosine, Euclidean (for continuous descriptors).

- **Role of Fingerprints and Features**
  - Fingerprints capture structural fragments or atom environments (e.g., ECFP, MACCS).
  - Descriptors encode physicochemical properties (e.g., logP, MW, H-bond donors).
  - The chosen representation defines what "similar" means in a given context.

- **Applications in Drug Design**
  - **Virtual screening:** find analogues of known actives.
  - **Scaffold hopping:** identify novel chemotypes with similar pharmacophoric patterns.
  - **Activity cliffs:** small structural changes → large activity differences.
  - Key tool for prioritizing compounds in early discovery stages.

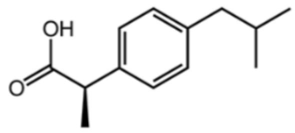
Similarity versus activity



Similarity versus activity. Three vascular endothelial growth factor receptor 2 ligands are shown that represent different (vertical vs horizontal) similarity–activity (potency) relationships.

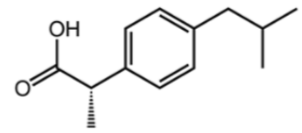
Complex similarity relationships

(R)-(-)-Ibuprofen



COX-1	COX-2	HSL
no	no	no

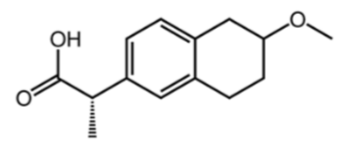
(S)-(+)-Ibuprofen



Bioavailability 49 -73%

COX-1	COX-2	HSL
yes	yes	no

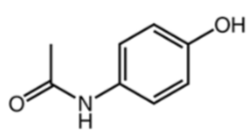
(S)-(+)-Naproxen



Bioavailability 95%

COX-1	COX-2	HSL
yes	yes	yes

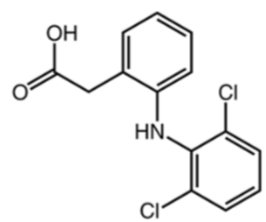
Paracetamol



Bioavailability ~100%

COX-1	COX-2	HSL
no	yes	no

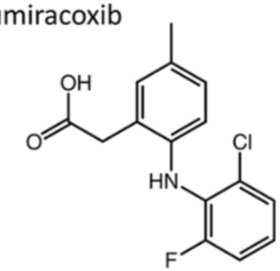
Diclofenac



Bioavailability >99%

COX-1	COX-2	HSL
yes	yes	no

Lumiracoxib

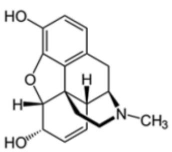


Bioavailability ~74%

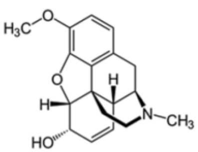
COX-1	COX-2	HSL
no	yes	no

Complex similarity relationships. Cyclooxygenase (COX) inhibitors and their activity profiles are compared. HSL – hormone sensitive lipase.

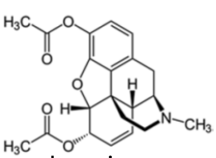
Measuring molecular similarity



morphine

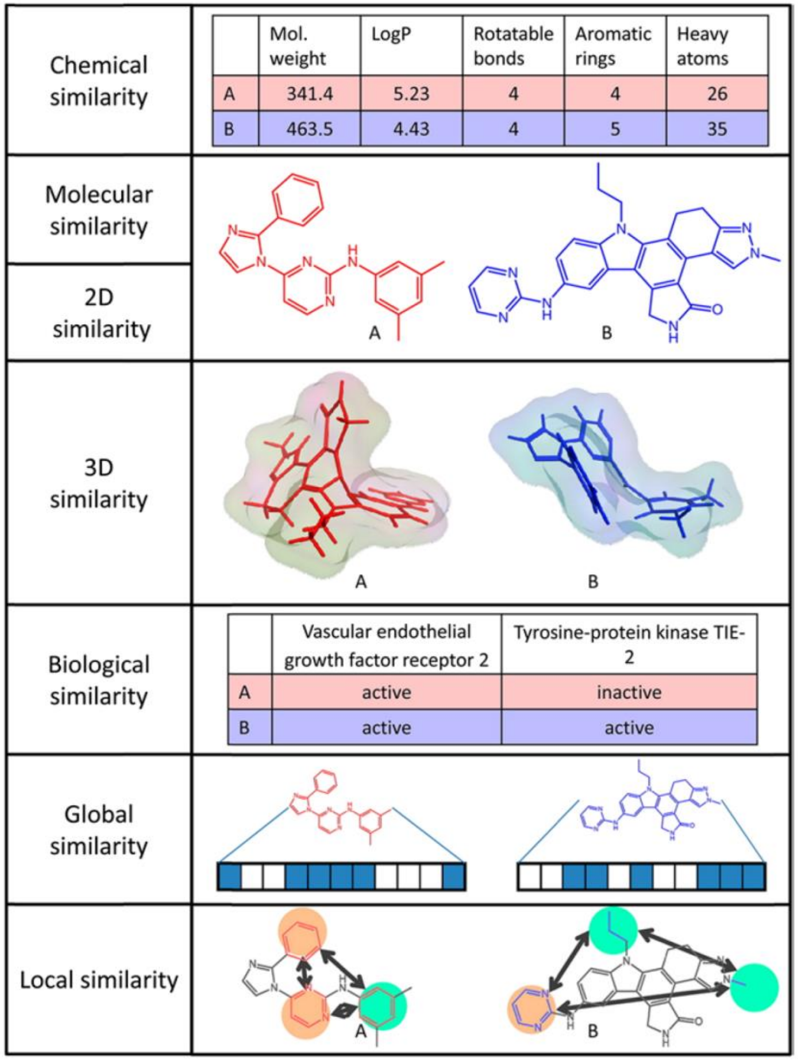


codeine



heroine

Similar structures similar functions

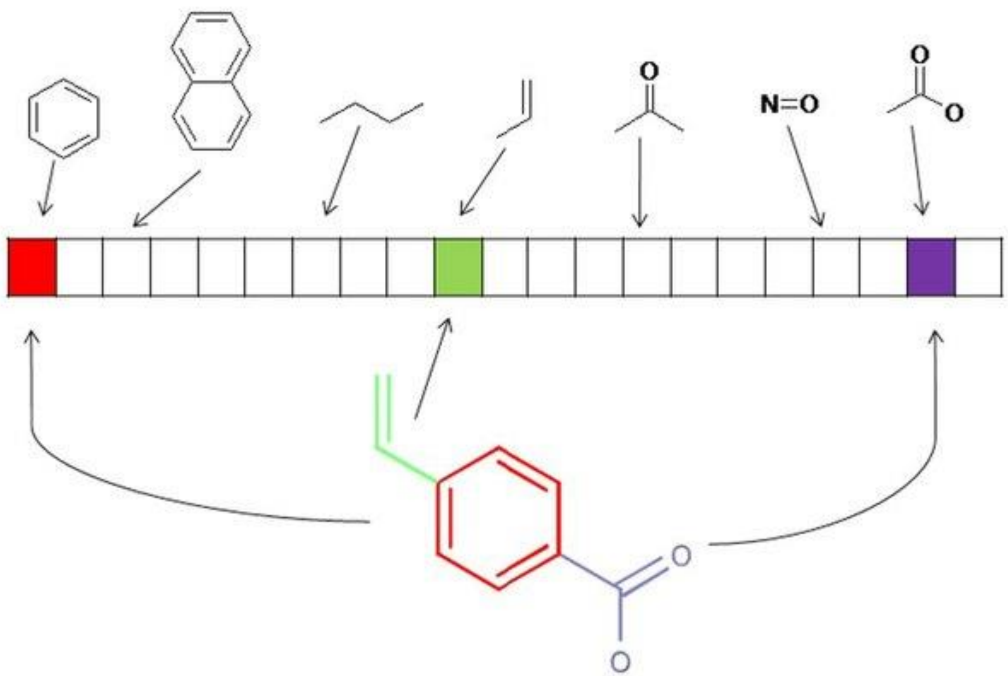


	Mol. weight	LogP	Rotatable bonds	Aromatic rings	Heavy atoms
A	341.4	5.23	4	4	26
B	463.5	4.43	4	5	35

	Vascular endothelial growth factor receptor 2	Tyrosine-protein kinase TIE-2
A	active	inactive
B	active	active

Molecular substructure fingerprint

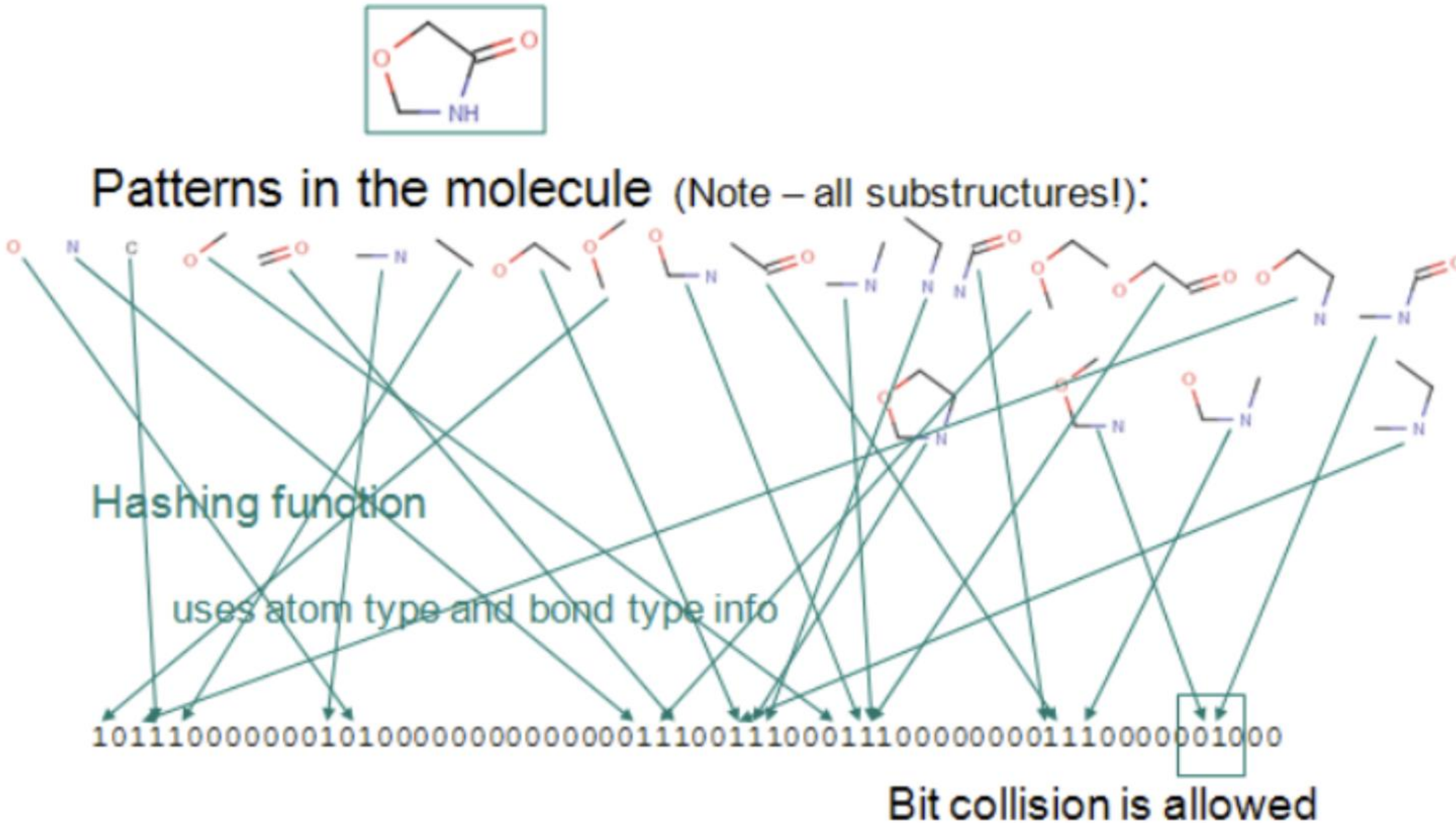
Representation of a molecular substructure fingerprint with a substructure fingerprint dictionary of given substructure patterns. This molecule is represented in a series of binary bits that represent the presence or absence of substructures in the molecules.



21-bit fingerprint

Fingerprint bit-vector: 100000000100000000010

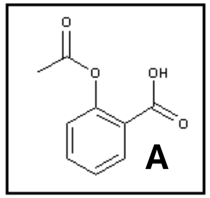
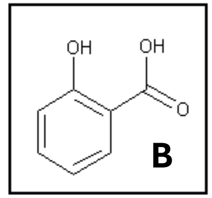
Hashed fingerprint

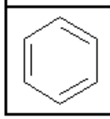
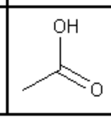
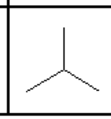
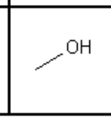
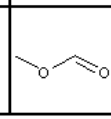
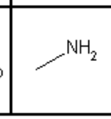
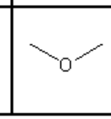
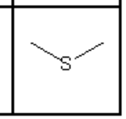


Common fingerprints

Fingerprint	Type / Representation	Typical Bit Length	Key Features	Notes / Use Cases
MACCS Keys	Structural key-based	166 bits	Fixed dictionary of predefined substructures (e.g., halogens, rings, bonds)	Simple and interpretable; limited coverage of chemical space
PubChem Fingerprint	Structural key-based	881 bits	Encodes presence of predefined SMARTS patterns	Used in PubChem database; good for database-wide similarity
Morgan (ECFP)	Circular (topological)	Variable (typically 1024 or 2048 bits)	Encodes atom environments within a given radius (e.g., radius=2 → ECFP4)	Widely used in ML; captures substructure diversity effectively
RDKit Fingerprint	Path-based	Variable (1024–2048 bits typical)	Hashes atom–bond paths up to a certain length	Fast and general-purpose; good for quick similarity screening
Atom Pair Fingerprint	Distance-based (topological)	Variable (2048+ bits typical)	Encodes atom types and topological distances	Sensitive to molecular shape; larger bit vectors
Topological Torsion Fingerprint	Path-based (torsions)	Variable	Encodes sequences of four bonded atoms (torsions)	Good descriptor of 3D-related connectivity patterns
Avalon Fingerprint	Hybrid (pattern + substructure)	512–2048 bits	Combines structural keys and path-based features	Compatible with RDKit; flexible for custom workflows
Pharmacophore Fingerprints (e.g., FCFP)	Feature-based circular	Variable	Encodes pharmacophoric features (H-bond donor, acceptor, aromatic, etc.)	Useful in ligand-based screening and QSAR models

Similarity with Tanimoto coefficient

A	**1**	**1**	**0**	**1**	**1**	**0**	**1**	**0**
B	**1**	**1**	**0**	**1**	**0**	**0**	**0**	**0**
								

$$\text{T.c.} = \frac{3}{5+3\text{-}3} = 0.6$$

$$\text{Tanimoto coefficient} = \frac{N_{\text{AB}}}{N_{\text{A}} + N_{\text{B}} - N_{\text{AB}}}$$

N_A – number of features of A

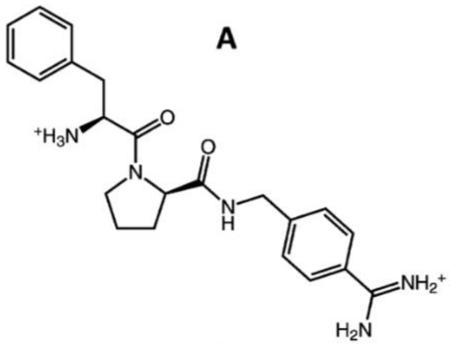
N_B – number of features of B

C_{AB} – common features of A and B

Two molecules may display a T.c of 1 and not be identical ! (different isomers, tautomers, etc...)

Different measures produce different results

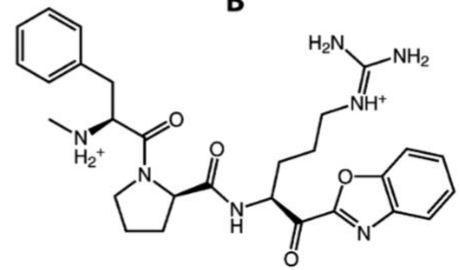
A



a

48 MACCS features
58 ECFP4 features

B



b

63 MACCS features
82 ECFP4 features

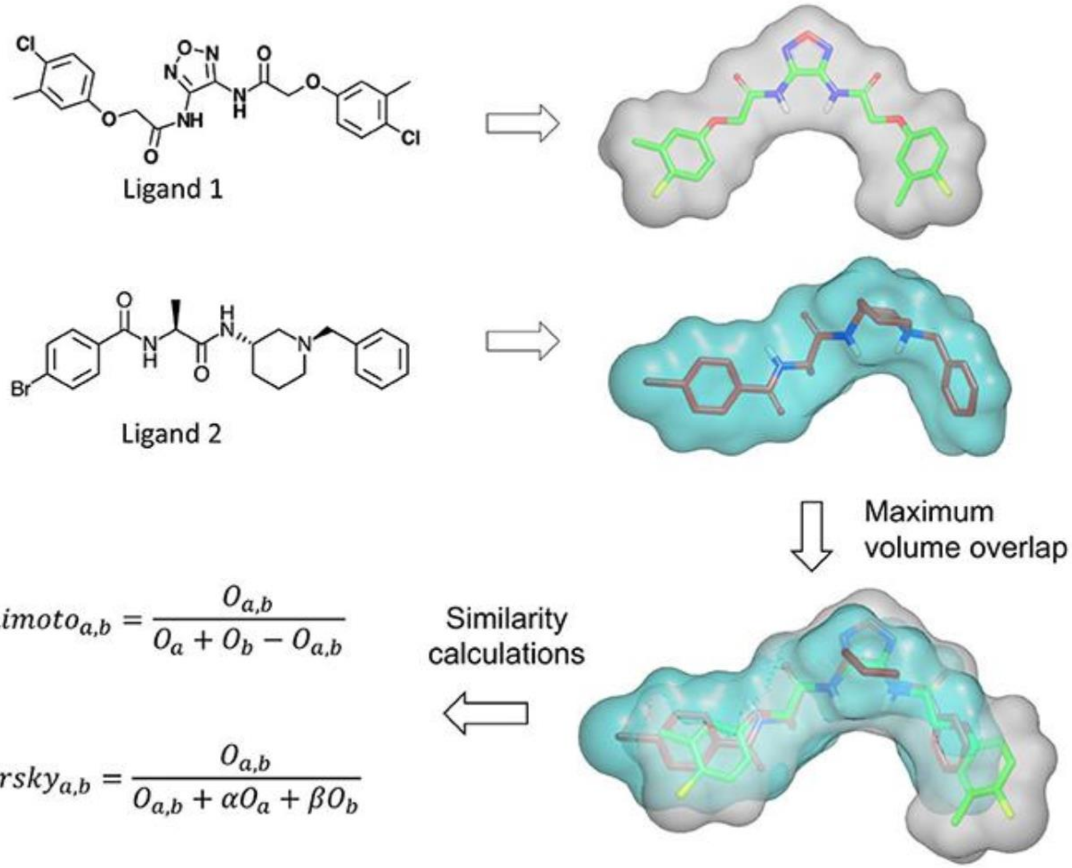
c

47 shared MACCS features
35 shared ECFP4 features

	$Tc(A,B)=\frac{c}{a+b-c}$	$Dc(A,B)=\frac{2c}{a+b}$	$Tv(A,B,\alpha)=\frac{c}{\alpha a+(1-\alpha)b}$ $\alpha=0.1$	$\alpha=0.9$
MACCS	0.73	0.85	0.76	0.95
ECFP4	0.33	0.50	0.44	0.58

Comparison of similarity coefficients. For two thrombin inhibitors Dice, Tanimoto, and Tversky coefficients are compared using MACCS and ECFP4. Tversky similarity calculations were carried out using different parameter settings.

3D similarity (molecular volume overlap)



$$Tanimoto_{a,b} = \frac{O_{a,b}}{O_a + O_b - O_{a,b}}$$

$$Tversky_{a,b} = \frac{O_{a,b}}{O_{a,b} + \alpha O_a + \beta O_b}$$